Perceptual Screen Content Image Hashing Using Adaptive Texture and Shape Features

Xue Yang, Ziqing Huang, Yonghua Zhang, Shuo Zhang, and Zhenjun Tang

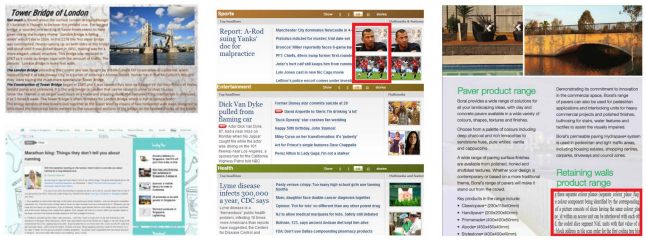
***Abstract*—With the flourishing development of multi-client interactive systems, a new type of digital image known as Screen Content Image (SCI) has emerged. Unlike traditional natural scene images, SCI encompasses various visual contents, including natural images, graphics, and text. Because the multi-region distribution characteristics of screen content images result in the presence of blank regions, malicious modifications are easier to operate and harder to perceive, making a serious threat to visual content security. To this end, this paper proposes a color screen content image hashing algorithm using adaptive text regions features and global shape features. Specifically, the text regions are adaptively collected by calculating the local standard deviation of sub-blocks. Then, quaternion Fourier significant maps are computed for the text regions, and texture statistical features are further extracted to reflect the essential visual content robustness. Moreover, the global shape features are represented from the entire color SCI to ensure the discrimination. Finally, the hash sequence with a length of 142 bits is derived from the above features. Importantly, a specialized tampering dataset for SCIs has been established, and the proposed hashing shows highly sensitive to malicious modifications with a satisfactory detection accuracy. Meanwhile, the ROC curve analysis indicates that the proposed method outperforms existing hashing algorithms.**

***Index Terms*—Screen content image, perceptual hashing, adaptive texture, global shape, tampering detection.**

I. Introduction

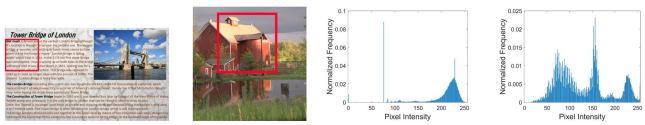
In THE era of the Internet of Things and 5G communication, multi-client interactive systems such as virtual exhibition halls, distance education, and e-magazines are rapidly evolving [1]. It is noteworthy that the visual content in these systems is no longer limited to a single natural scene image, and the content of current digital images has mixed sources, e.g. natural scenes, computer-generated graphics, text. Such images can be denoted as Screen Content Image (SCI) [2], [3]. Several SCIs are shown in Fig. 1, which has richer content with text and graphics in addition to traditional natural images.

In the interests of conforming to the requirements of information transmission and sharing in different scenarios, conventional operations (e.g. compression, brightness adjustment, scaling) are usually applied to SCIs [4]. However, it cannot be ignored that SCI is composed of text, images, graphics. Because its multi-region distribution characteristics result in the presence of blank regions, malicious tampering, forgery and other modifications are easier to operate and harder to perceive, making a serious threat to visual content security. The red box of Fig. 1(b) and (c) are maliciously added contents based on their region distribution characteristics, which are difficult to detect but greatly change the visual perception. The ever-increasing SCIs urgently require feasible content description schemes to solve their efficient storage and visual security issues.



(a) (b) (c)

Fig. 1. SCIs from the SCID [38] and SIQAD [39] database.



(a) (b) (c) (d)

Fig. 2. The normalized histograms of pixel distributions.

Inspired by visual content description schemes for natural images, perceptual hashing should be a method to address the content security issues of SCIs. The concept of perceptual hashing was first proposed at the International Conference on Image Processing (ICIP) in 1996 [5], achieving satisfactory authentication accuracy based on good robustness and discrimination [6], [7], [8], [14], [15], [16]. Unfortunately, there are significant differences between SCIs and natural images in terms of generation pathways, organizational structures, visual understanding, and statistical laws [9], [34], and previous perceptual hashing algorithms directly applied to SCIs have failed to achieve good performance [9], making it difficult to ensure security in various scenarios. Fig. 2 shows the normalized histograms of pixel distributions for natural and screen images, clearly highlighting the differences between the two. Therefore, it is essential to conduct specialized research on screen content image hashing to promote its positive development [10], [11].

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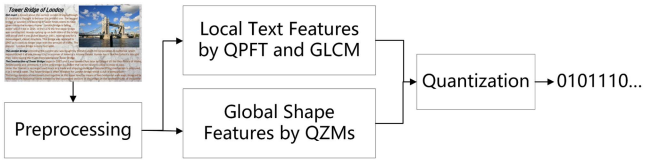


Fig. 3. The proposed perceptual hashing for SCIs.

SCIs exhibit unique multi-region content distribution characteristics, with significant differences in content expression, organization, and data statistics across different regions (such as images, text, tables, and graphics). These differences severely restrict effective description and encoding of visual content, presenting a key bottleneck in the research of perceptual hashing for screen images. Fortunately, existing studies have shown that text regions possess stronger visual expression compared to other regions [12], [13], and the acquisition of visual information primarily relies on eye movements within the text regions. Compared to low-level spatial information, extracting texture or color information provides higher accuracy [36]. Based on these facts, we focus on the texture features derived from text regions and global shape structure.

In detail, an innovative adaptive region segmentation algorithm is designed to obtain the text regions, and quaternion Fourier significant maps are computed for the text regions. Texture statistical features are further extracted to reflect the essential visual content robustness. Moreover, the global shape features are represented from the entire color SCI to ensure the discrimination. In summary, aiming at the multi-region content distribution characteristics of screen images, a perceptual hashing algorithm for color screen images that takes into account robustness and discrimination is proposed for solving its visual content security problem.

II. The Proposed Perceptual Hashing Method

Fig. 3 illustrates the framework of the proposed screen image hashing method, which mainly includes preprocessing, local texture feature extraction, global feature extraction, and compression quantization. The specific algorithmic flow will be discussed in detail in the following subsections.

A. Preprocessing

Considering the multi-region content distribution of SCIs, the human often first focuses on text areas through "fixation" and "saccade"[31], [32]. To effectively extract visual salient text regions, this paper proposes an adaptive region segmentation algorithm. The algorithm utilizes standard deviation to capture texture-rich areas. By calculating the local standard deviation and using the average as a threshold, it adaptively identifies text blocks and generates a text block index matrix for precise segmentation. This method ensures accurate recognition of text regions across different images, laying a solid foundation for subsequent feature extraction and processing.

B. Local Text Regions Saliency Feature Representation

1) Text Regions Saliency. PQFT: Through the operations in previous section, we locate the text regions from one color image based on text index matrix. However, the obtained text regions still exhibit much redundancy. To extract salient information and enhance visual robustness [33], the phase spectrum of the Quaternion Fourier Transform [18] is used to generate saliency map of the text regions. The steps regarding $PQFT$ are briefly explained as follows.

There is a "color-opponent component" system in the human brain, making it highly sensitive to color information. Based on this, the following color channels are obtained by simulating this color-opponent component system:

$$RG(i,j) = R(i,j) - G(i,j), \tag{1}$$

$$BY(i,j) = B(i,j) - Y(i,j), \tag{2}$$

where (i,j) is the position of each pixel in the spatial domain.

First, color screen image can be expressed in terms of the quaternion $q(i,j)$ through the following equation:

$$q(i,j) = RG(i,j)\mu_1 + BY(i,j)\mu_2 + I(i,j)\mu_3, \tag{3}$$

where $I(i,j)$ is the intensity channel.

Then, $q(i,j)$ can be expressed in Sin form:

$$q(i,j) = f_1(i,j) + f_2(i,j)\mu_2, \tag{4}$$

$$f_1(i,j) = RG(i,j)\mu_1, \tag{5}$$

$$f_2(i,j) = BY(i,j) + I(i,j)\mu_1. \tag{6}$$

Second, performing the Fourier transform on the quaternion $q(i,j)$ to obtain $Q(u,v)$ [19], as follows:

$$Q(u,v) = F_1(u,v) + F_2(u,v)\mu_2, \tag{7}$$

$$F_\varphi(u,v) = \frac{1}{\sqrt{MN}} \sum_{j=0}^{M-1} \sum_{i=0}^{N-1} e^{-\mu_1 2\pi((\frac{iu}{N})+(\frac{jv}{M}))} f_\varphi(i,j), \tag{8}$$

where $\varphi \in \{1,2\}$, M and N are the width and height of the image, (i,j) are the position of each pixel.

Third, the inverse quaternionic Fourier transform can be obtained by changing the sign of the exponents and summing over i and j :

$$f_\varphi(i,j) = \frac{1}{\sqrt{MN}} \sum_{j=0}^{M-1} \sum_{i=0}^{N-1} e^{-\mu_1 2\pi((\frac{iu}{N})+(\frac{iv}{M}))} F_\varphi(u,v). \tag{9}$$

To analyze and process the phase information conveniently, $Q(u,v)$ is represented in polar coordinates as follows:

$$Q(u,v) = ||Q(u,v)||e^{\xi\Phi(u,v)}, \tag{10}$$

where $||Q(u,v)||$ is the amplitude spectrum, Let $||Q(u,v)|| = 1$ so that only the phase spectrum information is retained in $Q(u,v)$, the inverse transformation is performed to obtain the time-domain representation $q'(i,j)$:

$$q'(i,j) = RG'(i,j)\mu_1 + BY'(i,j)\mu_2 + I'(i,j)\mu_3, \tag{11}$$

where, $RG'(i.j)$, $BY'(i,j)$ and $I'(i,j)$ is the reconstructed information of the quaternion.

Finally, the saliency map is defined as follows:

$$S(i,j) = ||q'(i,j)||^2. \tag{12}$$

2) Texture Feature Expression. GLCM: To enhance the robustness of the algorithm, this paper calculates the Gray Level Co-occurrence Matrix (GLCM) of the saliency map in four directions ($0°$, $45°$, $90°$, and $135°$) with distances ranging from 1 to 8. The mean and standard deviation of the statistical features are then computed to extract texture features, thereby providing a

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more comprehensive understanding of the relationship between local texture features and pixels [35]. Consequently, the resulting local texture feature sequence has a length of 64, denoted as $\mathbf{G}$, which represents the visual feature expression of the text regions.

C. Global Feature Representation

For screen images with multiple regional distributions, the overall shape structure has important guiding values for effective discrimination between different images. Therefore, quaternion Zernike moment is introduced to address the global shape characteristics of color screen images that contain a large number of blank areas [37].

1) Quaternion Zernike Moments: According to the quaternion representation of screen images in Section B, the Zernike moments can be computed:

$$\mathrm{Znm} = \frac{n+1}{\pi} \int_0^1 \int_{-\pi}^{\pi} V_{nm}(r,\theta) f(r,\theta) dr d\theta, |r| \leq 1. \quad (13)$$

Thus, a set of global shape features of size $\frac{(n+1)(n+2)}{2}$ is obtained, which is invariant to the rotation of the screen image. The magnitude coefficients of these features form the vector $\mathbf{Z}$, which is utilized to generate the hash of the screen image.

D. Hash Generation

The local features and global features are quantized into binary bits through their respective means. The final hash sequence $\mathbf{H} = [\mathbf{G}, \mathbf{Z}]$ are generated, as well as the size is 64 + 78 = 142 binary bits.

III. Experimental Result and Analysis

A. Data Settings

Robustness Analysis Dataset: We selected 60 original screen images from two popular screen image datasets, SCID [38] and SIQAD [39], and applied 10 content-preserving operations to generate 76 content-similar images [20].

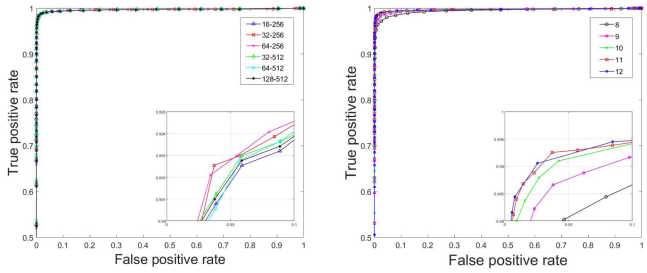
Discrimination Analysis Dataset: We collected 300 visually distinct screen images from online sources, including popular datasets, webpage screenshots, PowerPoint presentations, and magazine scans.

In addition, the parameters involved in our method are: image size of 256 × 256, block size of 32 × 32, and maximum order of 11 (i.e., $\mathbf{M} = 256$, $\mathbf{b} = 32$, order = 11). The similarity between hash sequences is measured by Hamming distance. We evaluated the algorithm's performance using three criteria: $\mathbf{P_{TPR}}$, $\mathbf{P_{FPR}}$, and Receiver Operating Characteristic (ROC).

B. Discussion of Key Parameters

1) Block Size Discussion: To obtain hash sequences of uniform length, we adjust the size of the input image using bilinear interpolation in the preprocessing step. $\mathbf{M}$ represents the image size and $\mathbf{b}$ indicates the block size. We tested several scenarios with $\mathbf{M}$ values of 256, 512 and $\mathbf{b}$ values of 16, 32, 64, 128. Fig. 4(a) displays the ROC curves, showing that the algorithm performs best with $\mathbf{M} = 256$ and $\mathbf{b} = 32$.

2) Maximum Order Discussion: Fig. 4(b) illustrates the ROC curves for different maximum orders in Zernike moments. It can be observed that 11 and 12 are closest to the top left corner, with AUC (Area under of curve) values of 0.99659 and 0.99304, respectively, indicating that the algorithm performs optimally when the maximum order is 11.



(a) (b)

Fig. 4. ROC curves among discussion of key parameters.

TABLE I
Statistical Results of Hamming Distances

Operation	Largest	Smallest	Mean	Deviation
Brightness adjustment	10	0	1.7292	2.0693
Contrast adjustment	5	0	0.84583	1.0416
Gamma correction	16	0	1.7583	2.0638
Gaussian low-pass filtering	51	1	17.0146	6.8438
Salt and pepper noise	25	0	1.015	2.3638
Speckle noise	21	0	1.4517	2.8905
JPEG compression	15	0	2.0333	2.334
Watermark embedding	19	0	1.16	2.464
Image scaling	8	0	1.0222	1.6871
Image rotation	47	1	15.2556	7.5207

C. Robustness Analysis

Content-similar image dataset are conducted to validate the robustness for the proposed scheme. Specifically, we calculated the Hamming distances between the hash sequences of the original images and the content-similar images. A smaller Hamming distance indicates better recognition capability and greater robustness. Table I presents the statistical results of the Hamming distances for each operation. The results show that the proposed method fully considers the multi-regional distribution characteristics of screen images. Through texture-driven feature extraction, it effectively preserves the visual content, meeting the requirements for perceptual robustness.

D. Discrimination Analysis

In the discrimination dataset, the Hamming distances between the hash codes of each pair of images were calculated, resulting in $C_{300}^2 = 44{,}850$ pairs. Fig. 5 presents the histogram of these Hamming distances, where the abscissa represents the Hamming distance and the ordinate represents the corresponding frequency. According to previous related experiments, when the Hamming distances follow a Gaussian distribution, it indicates that the algorithm has good discriminative ability [4]. Observed in the figure, the Hamming distances follow a Gaussian distribution, with a minimum value of 7, a maximum value of 71, and a mean of 39.4404. Compared to Table I in the robustness experiment, the statistical data of this experiment are significantly higher, further validating the superior discriminative performance of the algorithm.

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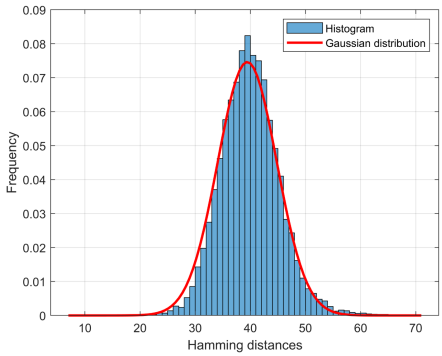


Fig. 5. Hamming distances distribution derived from different images.



Fig. 6. Examples of tampering image and corresponding hamming distances.

E. Tampering Sensitivity Analysis

Due to the multi-regional distribution characteristics of screen images and the presence of significant blank areas, they are more susceptible to malicious tampering that may not be easily perceptible. To verify the sensitivity of the proposed method to tampering with screen images, we created a specialized tampering detection dataset tailored for screen images, referred to as TSCID.[1] Most existing tampering datasets employ deep learning models for tampering, which may not align well with human visual perception and are therefore easier to detect. Consequently, we performed manual tampering, modifying portions of the images to conform to human visual perception and remain inconspicuous.

Regarding the original images from the SCID and SIQAD datasets, we classified tampering operations into two types: image region tampering and text region tampering. Each type includes three operations: copy [21], [22], [23], splice [24], [25], [26], and remove [27], [28], resulting in a total of $60 \times 6 = 360$ tampered images. Fig. 6 shows examples of tampered images, where the upper part is image tampering and the lower part is text tampering. We calculated Hamming distances between the original and tampered images, with results clearly presented between the Hamming distances of similar and different screen images. Table II displays the sensitivity test results for TSCID dataset, demonstrating that the proposed perceptual hashing performs well in sensitive to screen image tampering.

[1]TSCID: https://github.com/Yxx819/Tampered-Screen-Image-Dataset.

TABLE II
STATISTICAL RESULTS OF HAMMING DISTANCES

Operation		Largest	Smallest	Mean	Deviation
Image	Copy	36	0	13.4833	8.1188
	Splice	33	0	8.1617	7.1262
	Remove	36	2	14.4167	7.3445
Text	Copy	32	0	8.6167	8.2238
	Splice	37	0	6.65	6.6226
	Remove	11	0	1.6	2.5723

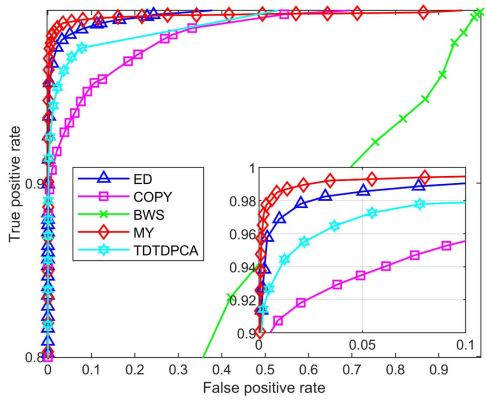


Fig. 7. ROC curves of different algorithms.

F. Comparison With Existing Methods

To demonstrate the superior performance of the proposed algorithm, we compared it with several existing perceptual hashing methods, including the BWS hashing method [29], ED hashing method [30], COPY hashing method [4], and 2D-2D PCA hashing method [17]. Fig. 7 shows the ROC curves of these algorithms, in which our method has achieved the best performance in robustness and discrimination. For a more intuitive comparison, we calculated the area under the ROC curve (AUC) for each method.The AUC values for the methods and our proposed algorithm are 0.99468, 0.98365, 0.83176, and 0.99659, respectively. It is evident that our perceptual hashing designed for screen images outperforms the others in classification performance, demonstrating strong classification capability.

IV. CONCLUSION

This paper presented an innovative perceptual hashing scheme specifically designed for screen images. Unlike natural scene images, screen images typically contain text, graphics, and both natural and computer-generated elements. Our proposed perceptual hashing algorithm for color screen images balances robustness and discrimination, introducing a texture feature description method based on quaternion Fourier transform specifically for text regions, significantly enhancing robustness. Experimental results demonstrate that this method effectively handles text regions and delivers excellent performance. Additionally, we developed a quaternion-based Zernike moment feature description method to address the multi-content nature of screen images, further improving the discrimination of the hashing algorithm. Overall, the proposed color screen image hashing scheme, based on human visual system perception, exhibits outstanding classification performance, with experimental results significantly outperforming other popular algorithms, particularly in handling text and multi-content regions.

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